IDENTIFICATION OF FILE EXTENSION MISMATCHES AND HIDDEN CONTAINER OBJECTS IN PROJECT FALCON

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Introduction

This forensic report details the analysis, integrity validation, and hidden object recovery operations performed on the evidence file **HIVE-FSA-003 Project Falcon.doc**. The major purpose of this study is to resolve a detected system extension mismatch, examine low-level file structure indications, and safely isolate any data contained inside the file container.

Tools

1. **HIVE File Header Analysis & Signature Intelligence (v2.1.0, build 2026-08-15)**: Utilized for automated extension verification, entropy calculations, and pattern-matching signature hunts.
2. **HxD Hex Editor (64-bit)**: Used as a stand-alone tool to manually examine the raw byte array at structural origin points and confirm results outside of the automated suite.
3. **Windows File Explorer & Photos App**: Used to confirm the programmatic execution of recovered visual artifacts and maintain isolated duplicate copies of evidence.

Scope and evidence

Ref.	Item	Size	SHA-256
FHAS-001	HIVE-FSA-003 Project Falcon.doc (original evidence)	1.44 MB	0a66327071f197fde24fa17f9d0a789efcaea335c5d4744db67282f77c9bd2f3
FHAS-002	Same file renamed HIVE-FSA-003 Project Falcon.zip	1.44 MB	0a66327071f197fde24fa17f9d0a789efcaea335c5d4744db67282f77c9bd2f3 (identical)
FHAS-003	Internal object, claimed .dat, true type PNG; recovered as final_artifact_09.png	1.38 MB	48617b2e93344fcb9c10c2c9de0d9ca0a20391ca7ed08d899b1ae9de552eb651

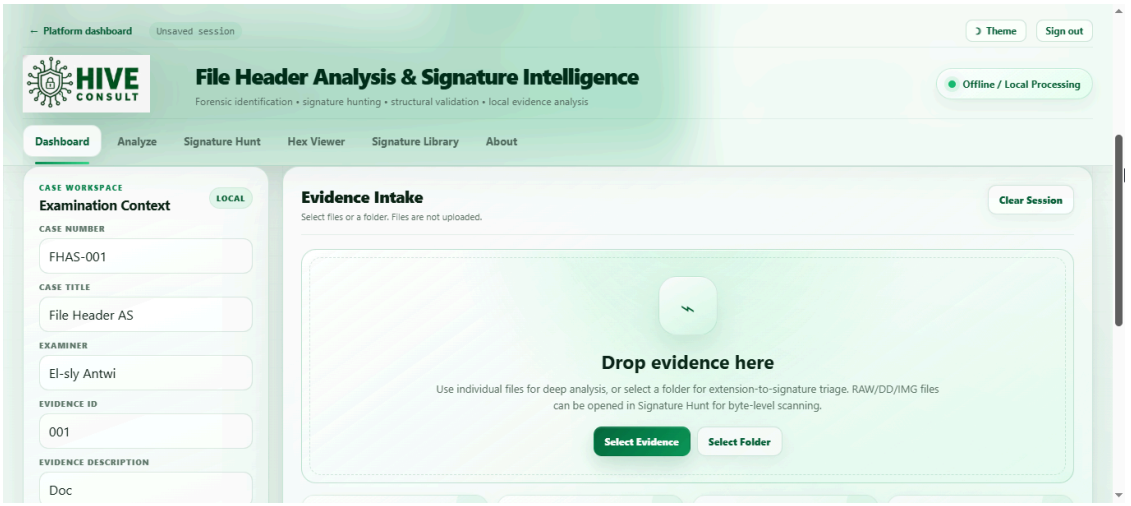
Methodology

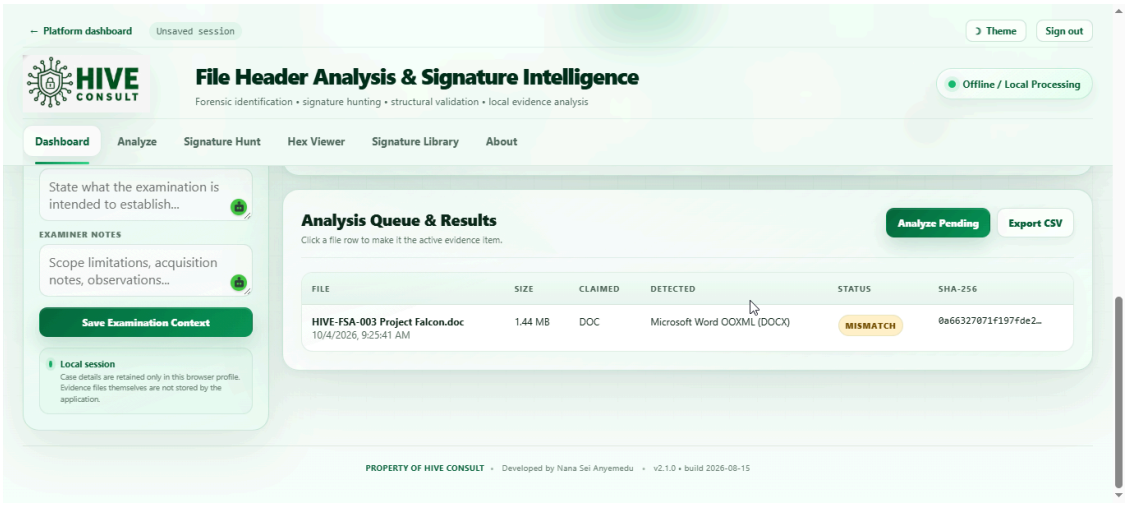
1. Record the claimed type from the file extension.
2. Load the file in HIVE, compute SHA-256 and entropy, and read the automated type detection.
3. Inspect the first bytes at offset 0x00000000 in the HIVE Hex Viewer and independently in HxD.
4. Compare the header to the signature library and search for the legacy OLE2 signature.
5. Hunt for further ZIP local file headers (50 4B 03 04 14) to find embedded entries.
6. Rename the file to .zip, confirm the hash is unchanged, and extract the contents.
7. Identify the concealed object by its header, restore the correct extension, open it, and record the flag.

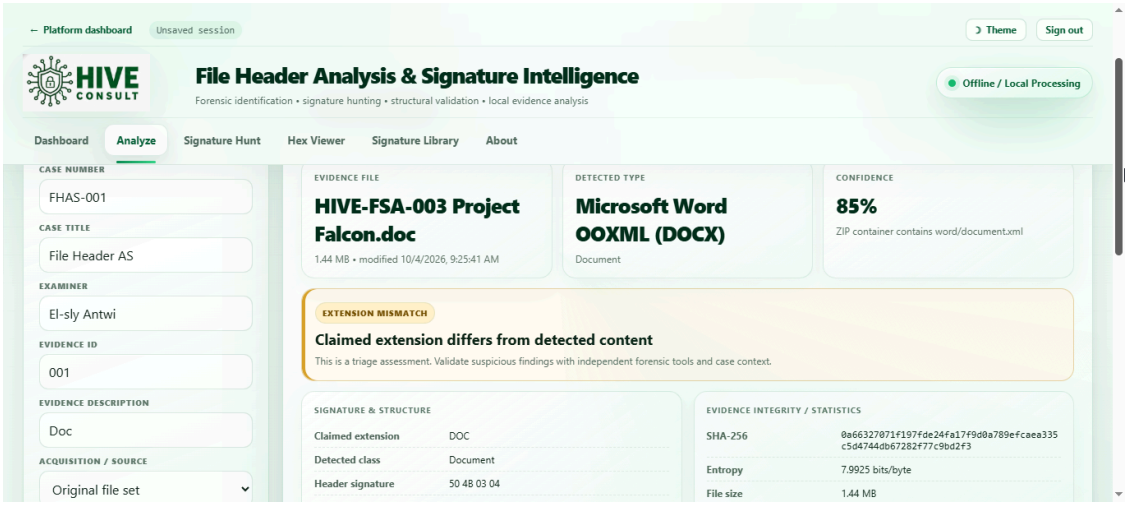
Findings

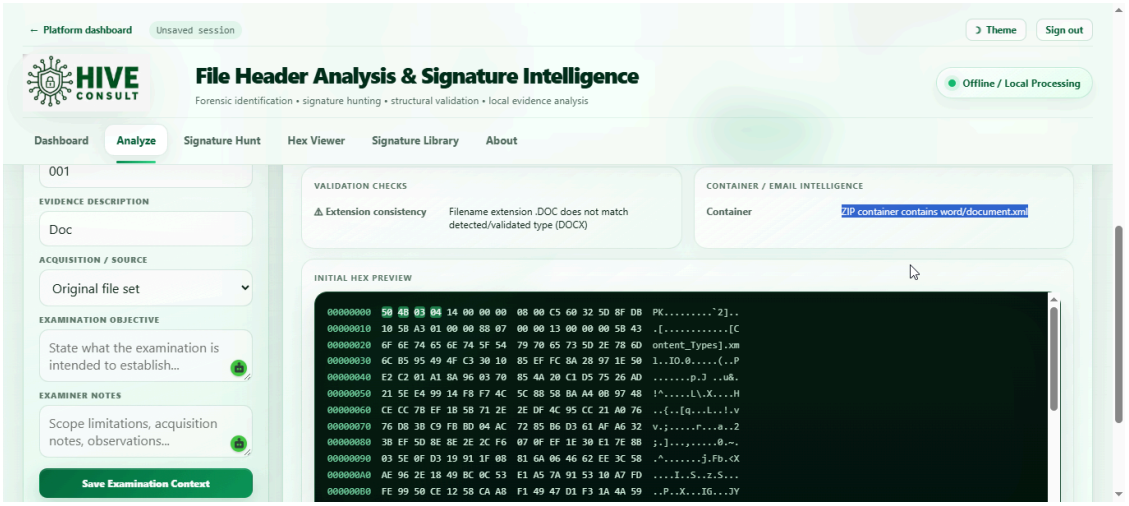
1. Claimed type and automated triage

The evidence file has the extension .doc, which claims a legacy Microsoft Word 97-2003 document. HIVE records the claimed extension as DOC but detects the content as Microsoft Word OOXML (DOCX) with 85% confidence, based on a ZIP container holding **word/document.xml.**



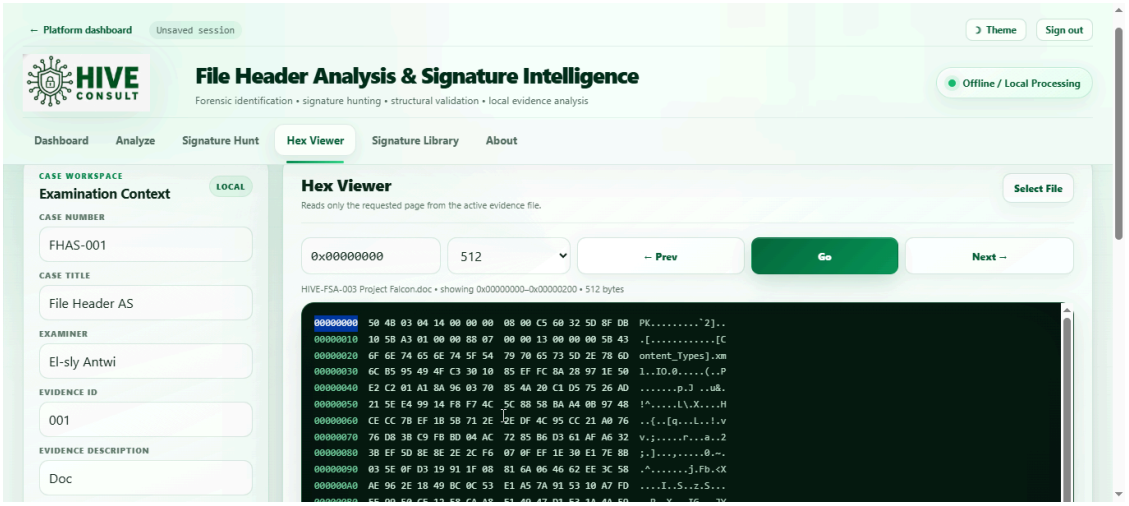


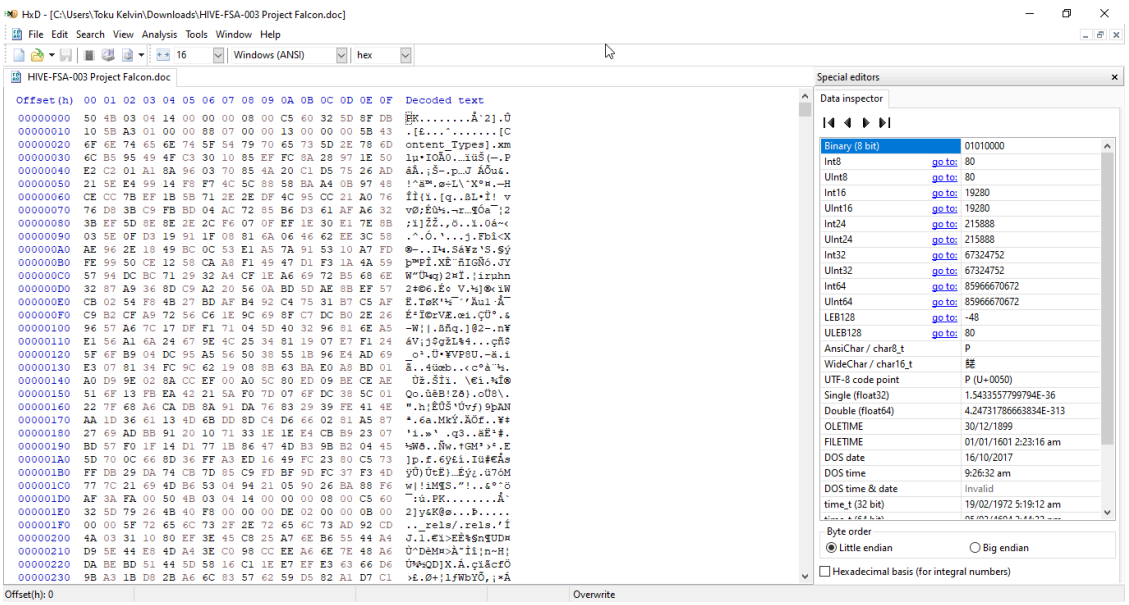


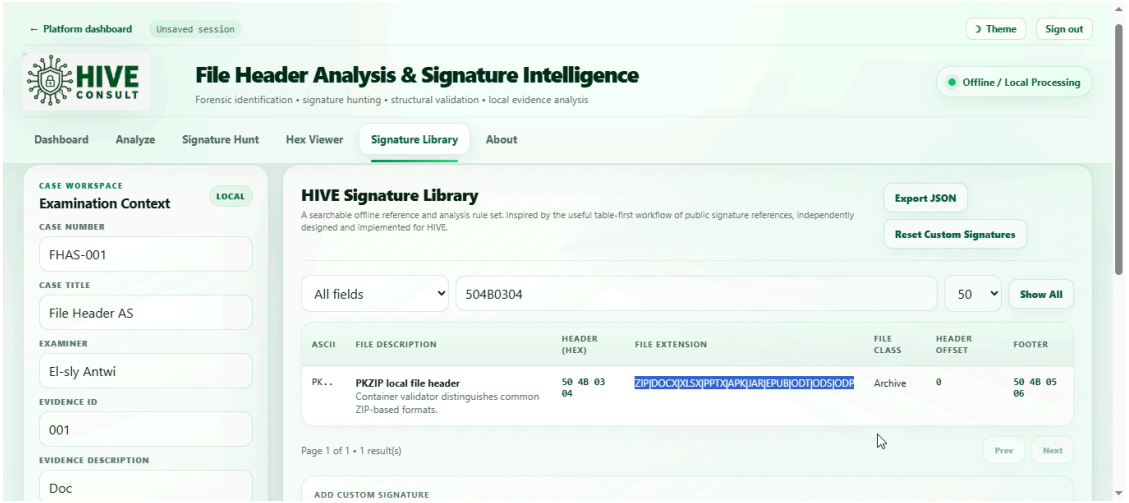


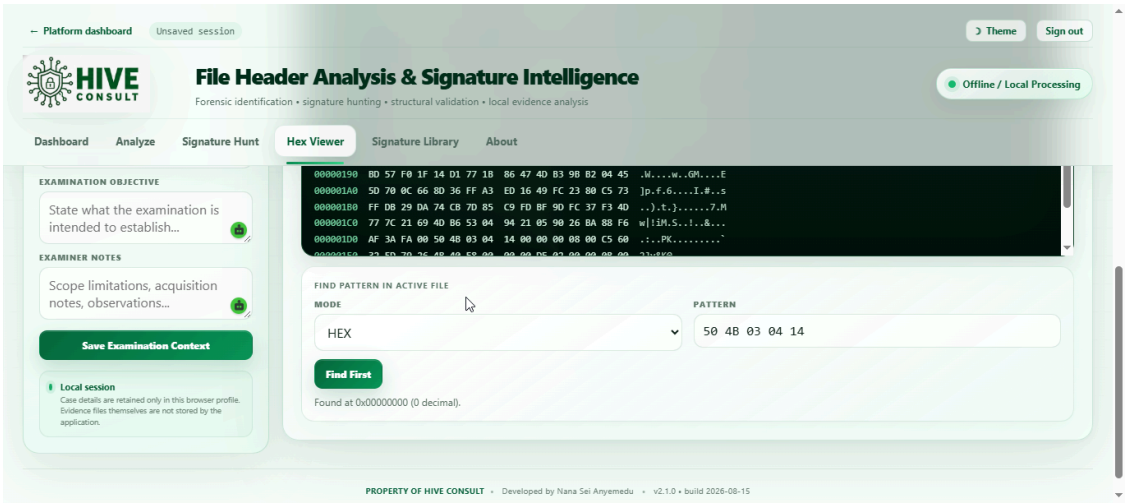
2. Header Examination

At offset 0x00000000 the file begins 50 4B 03 04 14 00 00 00 08 00. The decoded text reads PK, and [Content_Types].xml is legible beginning at offset 0x1E. HxD shows the same bytes and also shows _rels/.rels further in. The signature library lists 50 4B 03 04 as the PKZIP local file header, used by ZIP, DOCX, XLSX, PPTX and related formats. A search for 50 4B 03 04 14 finds a match at offset 0.


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LOCAL
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CASE TITLE
File Header AS
EXAMINER
El-sly Antwi
EVIDENCE ID
001
EVIDENCE DESCRIPTION
Doc
Hex Viewer
Reads only the requested page from the active evidence file.
Select File
0x00000000
512
← Prev
Go
Next →
HIVE-FSA-003 Project Falcon.doc • showing 0x00000000–0x00000200 • 512 bytes

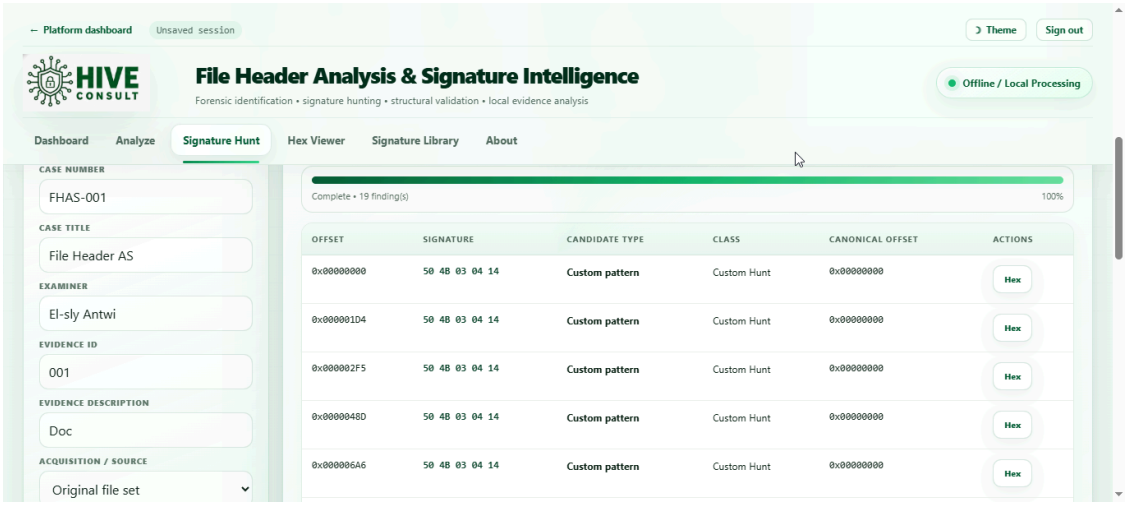

HxD - [C:\Users\Toku Kelvin\Downloads\HIVE-FSA-003 Project Falcon.doc]
File Edit Search View Analysis Tools Window Help
HIVE-FSA-003 Project Falcon.doc
Offset(h) 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F Decoded text
Special editors
Data inspector
Offset(h): 0
Overwrite

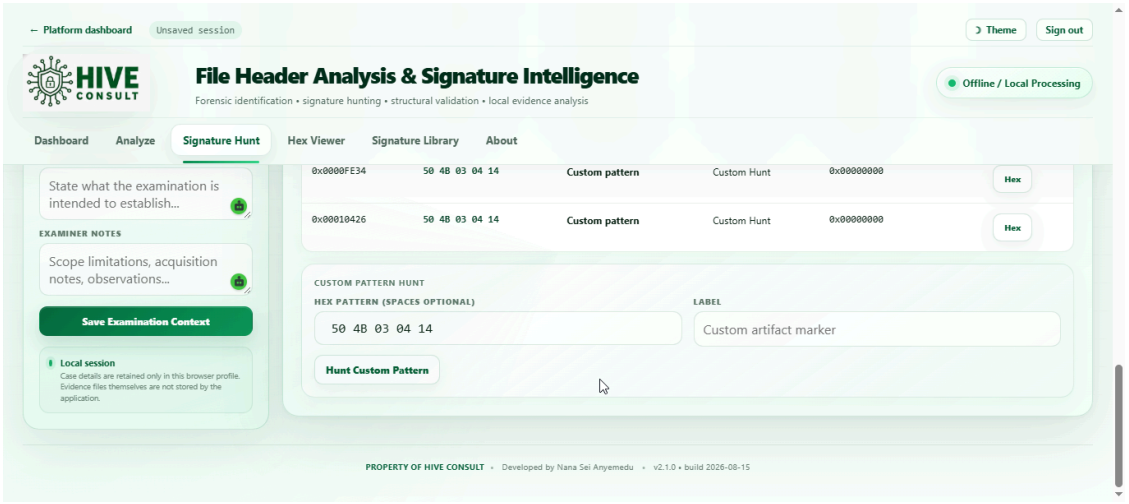

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File Header AS
EXAMINER
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EVIDENCE ID
001
EVIDENCE DESCRIPTION
Doc
HIVE Signature Library
A searchable offline reference and analysis rule set. Inspired by the useful table-first workflow of public signature references, independently designed and implemented for HIVE.
Export JSON
Reset Custom Signatures
All fields
504B0304
50
Show All
ASCII
FILE DESCRIPTION
HEADER (HEX)
FILE EXTENSION
FILE CLASS
HEADER OFFSET
FOOTER
PK..
PKZIP local file header
Container validator distinguishes common ZIP-based formats.
50 4B 03 04
ZIP|DOCX|XLSX|PPTX|APK|JAR|EPUB|ODT|ODS|ODP
Archive
0
50 4B 05 06
Page 1 of 1 • 1 result(s)
Prev
Next
ADD CUSTOM SIGNATURE

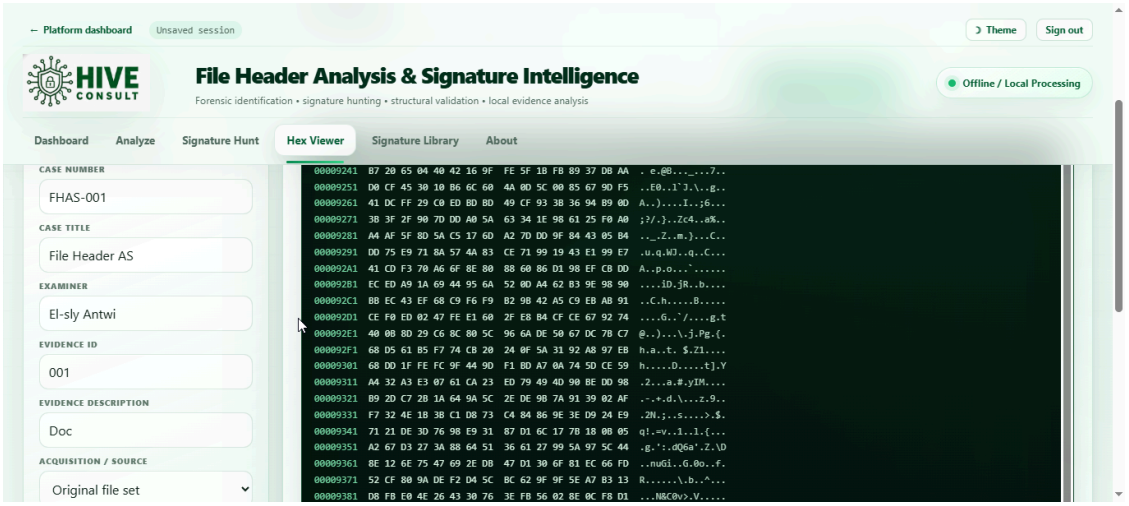


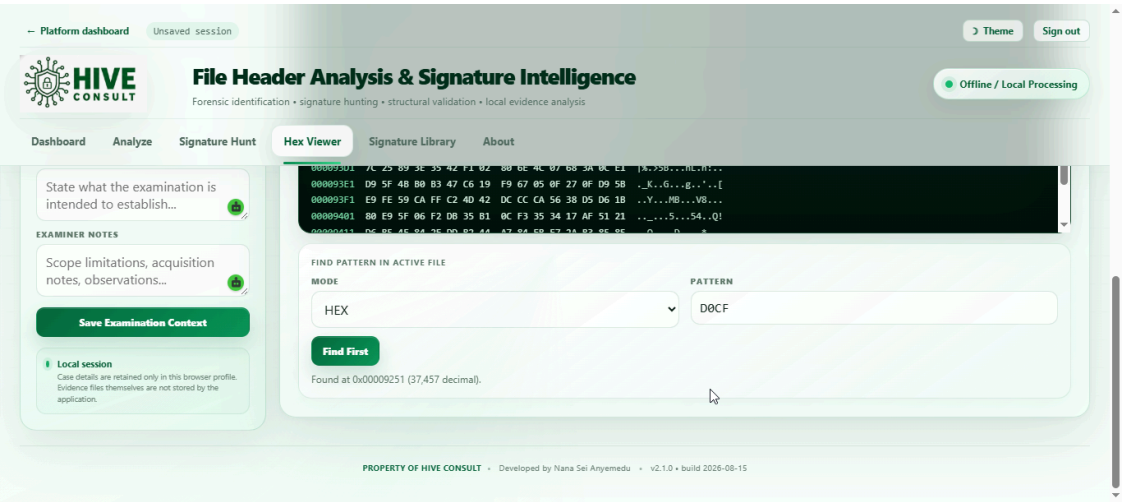
3. Internal structure and legacy .doc check

A legacy .doc is an OLE2 compound file and must begin with the 8 bytes D0 CF 11 E0 A1 B1 1A E1. The file begins with PK, so it is not an OLE2 file. A signature hunt for 50 4B 03 04 14 returned 19 matches. The first is at 0x00000000, which is the start of the container, and later ones begin at 0x000001D4, 0x000002F5, 0x0000048D and 0x000006A6. The last two displayed are at 0x0000FE34 and 0x00010426. These are consistent with multiple entries inside a ZIP archive


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File Header AS
EXAMINER
El-sly Antwi
EVIDENCE ID
001
EVIDENCE DESCRIPTION
Doc
ACQUISITION / SOURCE
Original file set
Complete • 19 finding(s)
100%
OFFSET
SIGNATURE
CANDIDATE TYPE
CLASS
CANONICAL OFFSET
ACTIONS
0x00000000 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
0x000001D4 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
0x000002F5 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
0x0000048D 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
0x000006A6 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex


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0x0000FE34 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
0x00010426 50 4B 03 04 14 Custom pattern Custom Hunt 0x00000000 Hex
State what the examination is intended to establish...
EXAMINER NOTES
Scope limitations, acquisition notes, observations...
Save Examination Context
Local session
Case details are retained only in this browser profile. Evidence files themselves are not stored by the application.
CUSTOM PATTERN HUNT
HEX PATTERN (SPACES OPTIONAL)
50 4B 03 04 14
LABEL
Custom artifact marker
Hunt Custom Pattern
PROPERTY OF HIVE CONSULT • Developed by Nana Sei Anyemedu • v2.1.0 • build 2026-08-15

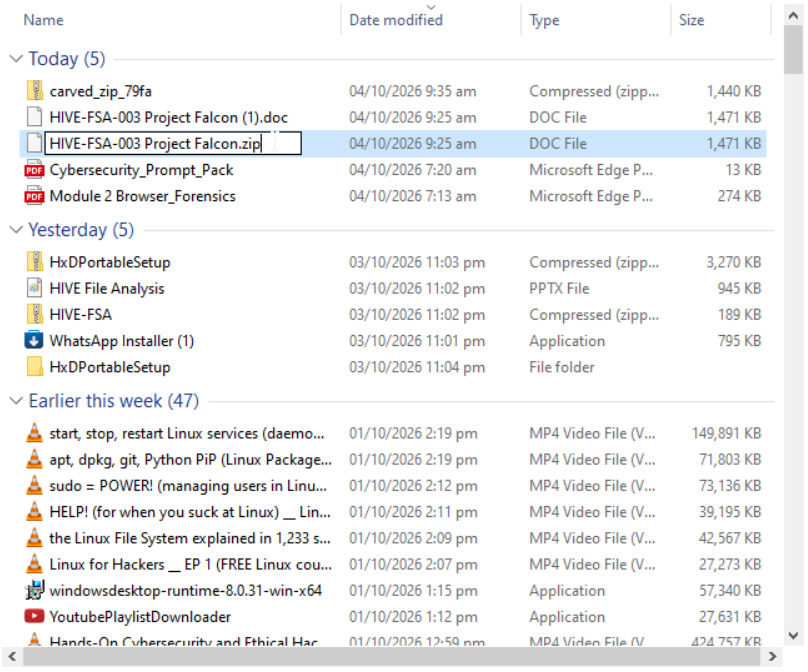

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EVIDENCE ID
001
EVIDENCE DESCRIPTION
Doc
ACQUISITION / SOURCE
Original file set
00009241 B7 20 65 04 40 42 16 9F FE 5F 1B FB 89 37 DB AA . e.@B..._...7..
00009251 D0 CF 45 30 10 B6 6C 60 4A 0D 5C 00 85 67 9D F5 ..E0..l`J.\..g..
00009261 41 DC FF 29 C0 ED BD 8D 49 CF 93 3B 36 94 B9 0D A..)....I..;6...
00009271 3B 3F 2F 90 7D DD A0 5A 63 34 1E 98 61 25 F0 A0 ;?/.}..Zc4..a%..
00009281 A4 AF 5F 8D 5A C5 17 6D A2 7D DD 9F 84 43 05 B4 .._.Z..m.}...C..
00009291 DD 75 E9 71 8A 57 4A 83 CE 71 99 19 43 E1 99 E7 .u.q.WJ..q..C...
000092A1 41 CD F3 70 A6 6F 8E B0 88 60 86 D1 98 EF CB DD A..p.o...`......
000092B1 EC ED A9 1A 69 44 95 6A 52 0D A4 62 83 9E 98 90 ....iD.jR..b....
000092C1 BB EC 43 EF 68 C9 F6 F9 B2 98 42 A5 C9 EB AB 91 ..C.h.....B.....
000092D1 CE F0 ED 02 47 FE E1 60 2F E8 B4 CF CE 67 92 74 ....G..`/....g.t
000092E1 40 0B 8D 29 C6 8C 80 5C 96 6A DE 50 67 DC 7B C7 @..)...\.j.Pg.{.
000092F1 68 D5 61 B5 F7 74 CB 20 24 0F 5A 31 92 A8 97 EB h.a..t. $.Z1....
00009301 68 DD 1F FE FC 9F 44 9D F1 BD A7 0A 74 5D CE 59 h.....D.....t].Y
00009311 A4 32 A3 E3 07 61 CA 23 ED 79 49 4D 90 BE DD 98 .2...a.#.yIM....
00009321 B9 2D C7 2B 1A 64 9A 5C 2E DE 9B 7A 91 39 02 AF .-.+.d.\...z.9..
00009331 F7 32 4E 1B 3B C1 D8 73 C4 84 86 9E 3E D9 24 E9 .2N.;..s....>.$.
00009341 71 21 DE 3D 76 98 E9 31 87 D1 6C 17 7B 18 0B 05 q!.=v..1..l.{...
00009351 A2 67 D3 27 3A 88 64 51 36 61 27 99 5A 97 5C 44 .g.':.dQ6a'.Z.\D
00009361 8E 12 6E 75 47 69 2E DB 47 D1 30 6F 81 EC 66 FD ..nuGi..G.0o..f.
00009371 52 CF 80 9A DE F2 D4 5C BC 62 9F 9F 5E A7 B3 13 R......\.b..^...
00009381 D8 FB E0 4E 26 43 30 76 3E FB 56 02 8E 0C F8 D1 ...N&C0v>.V.....



The file is not a legacy .doc. It is a ZIP-based Microsoft Word OOXML (.docx) document with the wrong extension.

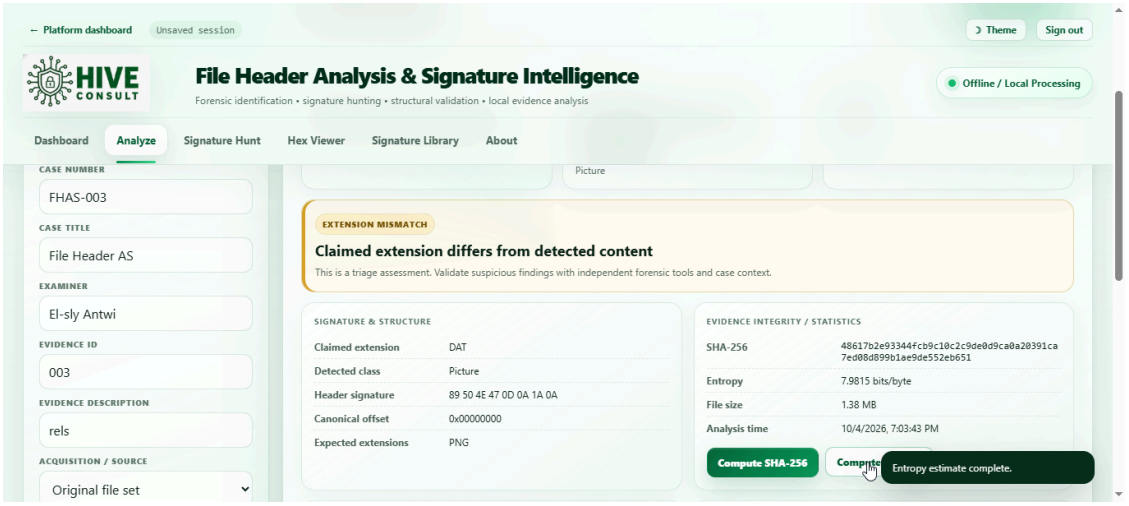
4. Extension correction and integrity

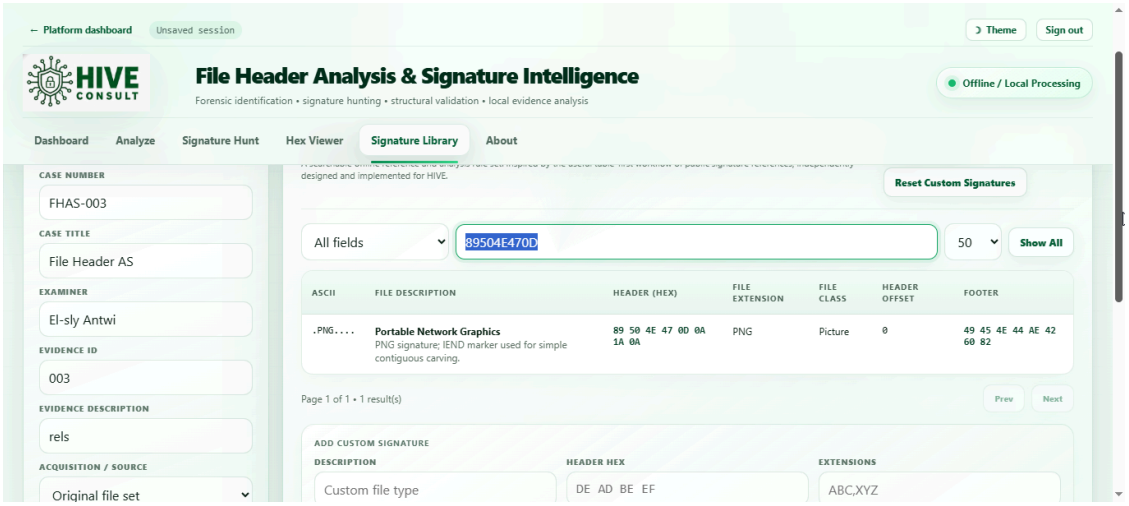
The file was renamed to HIVE-FSA-003 Project Falcon.zip so that it could be opened as an archive. HIVE shows the same header bytes, the same modified time (10/4/2026, 9:25:41 AM) and the same SHA-256 as before, which confirms the rename did not alter the content.



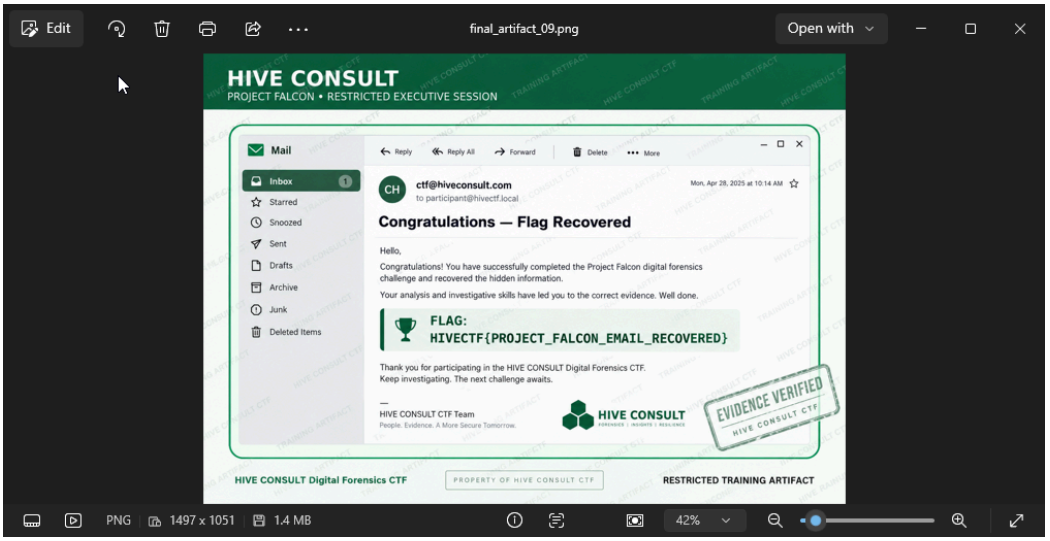
5. Concealed object and flag

So after extracting the zip, I ran a file analysis on the extracted files from the zip. Not all were consistent; there were some mismatches, so I decided to correct them to their right extensions. A file extracted from the archive final_artifact_09 .dat claimed extension DAT, detected class Picture, header signature 89 50 4E 47 0D 0A 1A 0A at offset 0x00000000, expected extension PNG, size 1.38 MB. It raises an extension-mismatch warning. The signature library identifies this header as Portable Network Graphics.





The correct extension (.png) was restored, and the file opened in the Windows Photos app. It is final_artifact_09.png, 1497 × 1051 pixels, 1.4 MB. The flag



6. Why the object is invisible in Microsoft Word

Word shows material.xml locations on the page, together with the sections that the document's relationship files relate to. The PNG is kept in the package under a .dat extension, so it is not an ordinary embedded picture, and Word has no menu that lists the raw portions of the package. As a result, just the document text is visible to a typical user that opens the file. Despite the .doc extension, Word opens the file without any issues since it determines the true format based on the content rather than the file name.

7. Forensic conclusion

HIVE-FSA-003 Project Falcon.doc is an evidence file that claims to be an old Word 97-2003 document. Nevertheless, its first magic bytes (50 4B 03 04) indicate a compressed ZIP container without any OLE2 signature at all. Structural parsing of word/document.xml and [Content_Types].xml verifies that it is an OOXML (.docx) asset concealed by the wrong extension wrap. When the container was disassembled, a secret internal item (.dat) containing magic bytes that matched a real PNG image was discovered upon deconstruction of the container. The token flag HIVECTF{PROJECT_FALCON_EMAIL_RECOVERED} was successfully revealed by restoring the true extension. Normal users are unable to access this graphic since it is not programmatically connected within the page body code. Verified by HIVE and HxD, this research reveals that external extensions are unreliable indications, and forensic identification must depend exclusively on raw header signatures